

Use Case Table Presentation

Use Case Name, Actors & Scenario	Steps	Test Cases
Use Case Name: Search Bus by Route Actors: User Scenario: Users can search for a bus by entering the start and destination points.	1. User enters the start point and destination 2. User clicks the 'Search' button 3. System prepares a list of buses which operate on the efficient routes where efficient routes have been analyzed by the system 4. Displays the list	• Search Bus with Valid Route • Search Bus with Invalid Route • Attempt to View Details of a Bus with No Information • Search with Only Start Point Provided • Search with Only Destination Provided • Search for Past Date or Time • Search with Non-Existent Route
Use Case Name: View Nearby Bus Stops Actors: User, System Scenario: User wants to find bus stops nearby to start their journey.	1. User opens the application and logs in. 2. System accesses the user's current location. 3. User selects a destination. 4. System identifies bus stops within a predefined radius (e.g., 1km) of the user's location. 5. System displays a list of nearby bus stops with distances and estimated time to reach.	• User inputs a valid destination • User inputs an invalid destination
Use Case Name: Get Estimated Arrival Time for Nearby Buses Actors: User, System Scenario: User wants to know the estimated time to reach each nearby bus stop and the buses arriving soon.	1. User opens the application and logs in. 2. System accesses the user's current location. 3. User selects a destination. 4. System identifies bus stops within a predefined radius (e.g., 1km) of the user's location. 5. User selects a nearby bus stop from the list. 6. System calculates the user's estimated walking time to reach the bus stop. 7. System retrieves real-	• User views estimated arrival time for buses • No real-time bus data available

	time data of buses arriving at the selected bus stop. 8. System displays the list of buses with their estimated arrival times and the user's estimated time to reach the stop.	
Use Case Name: View Available Buses and Routes Actors: User, System Scenario: User wants to check available buses and routes to their destination from a nearby bus stop.	1. User selects a bus stop from the nearby list. 2. System shows a list of buses available from the selected stop that can reach the destination. 3. System displays route information, including stops along the way and the expected travel time.	• User checks available buses and routes • No buses available for selected route
Use Case Name: Show Most Efficient Routes for Source and Destination Actors: User Scenario: The system displays the most efficient bus routes between a given source and destination, considering factors such as travel time, bus frequency, and route length.	1. User enters a source and destination 2. User clicks the 'Find Routes' button 3. System analyzes routes based on travel time, frequency, and route length 4. System displays a list of efficient routes, with the top option highlighted	• Display Most Efficient Route for Valid Source and Destination • No Routes Available for Selected Source and Destination • Display Multiple Efficient Routes Sorted by Criteria • Efficient Route Display with Limited Bus Frequency • Route Display with User-Preferred Efficiency Criteria
Use Case Name: Track a Bus Actors: User, API Scenario: User searches a bus or gives ID of a bus, API fetches the current location data of the bus and the data is displayed to the user	1. User goes to the "Track Live Location" page 2. User searches for a bus with route or name and time, or enter the bus ID directly 3. System fetches the location data of the bus from API 4. System shows the live location of the bus through a map	• Successful Bus Tracking by ID • Successful Bus Tracking by Searching the Bus • Invalid Bus ID Entry • No Results Found While Searching Bus • API Failure/Timeout

Use Case Name: Monitor Bus Speed and Notify Admin on Speed Limit Violation Actors: System, API, Admin Scenario: The system automatically monitors bus speeds in real-time, and if a bus exceeds the predefined speed limit, a notification is immediately sent to the admin.	1. System continuously monitors the live speed of each bus. 2. System compares the bus's current speed with its predefined speed limit. 3. If a bus is detected exceeding its speed limit, the system flags this as a violation. 4. System sends an immediate notification to the admin, including the bus ID, current speed, and timestamp of the violation.	• Speed Limit Violation Detection and Notification Sent to Admin • Bus Within Speed Limit - No Notification Sent • System Response to Multiple Speed Violations • System Monitoring Bus Speeds in Real-Time
Use Case Name: Submit a Complaint Actors: User, System Scenario: A user encounters an issue with a bus service and wants to lodge a formal complaint	1. The user navigates to the 'Submit Complaint' section. 2. The user selects the bus route, bus number, or bus name related to the complaint. 3. The user fills in details of the complaint, such as time, date, issue type (e.g., delays, cleanliness, driver behavior), and a description. 4. The user submits the complaint. 5. The system acknowledges the submission and assigns a unique complaint ID.	• Submit Complaint with Valid Data • Submit Complaint with Invalid Bus Service ID. • Submit Complaint with Missing Fields. • Submit Valid Media File Type. • Submit Invalid Media File Type. • Filing a Duplicate Complaint.
Use Case Name: View Complaints Actors: Admin Scenario: An admin wants to view a list of complaints submitted by users regarding various bus services.	1. The admin navigates to the Complaints Management section. 2. The admin selects View All Complaints. 3. The user fills in details of the complaint, such as time, date, issue type (e.g., delays, cleanliness, driver behavior), and a description. 4. The system displays a list of complaints, showing details like complaint ID, bus route, date, and	• Retrieving Complaints for a Specific Service. • Filtering Complaints by Status/Time • Filtering Complaints Using Invalid Data for Filtering. • Viewing Detailed Information of a Complaint

	complaint status. 5. The admin filters or sorts the complaints as needed.	
Use Case Name: Assign Complaint to Support Team Actors: Admin, Support Team Scenario: An admin assigns a complaint to a relevant support team for resolution.	1. The admin navigates to the list of complaints. 2. The admin views the details of a specific complaint. 3. The admin selects the option to assign the complaint. 4. The admin chooses a support team or agent from the list. 5. The system assigns the complaint to the selected team or agent and notifies them.	• Processing a Complaint.